Nixor O Levels
Test Week 1
Biology
Grade 10

34 + 2

36

Student Nam _____

Student ID: _____

Section : _____

Time: 50 minutes

Max marks: 40

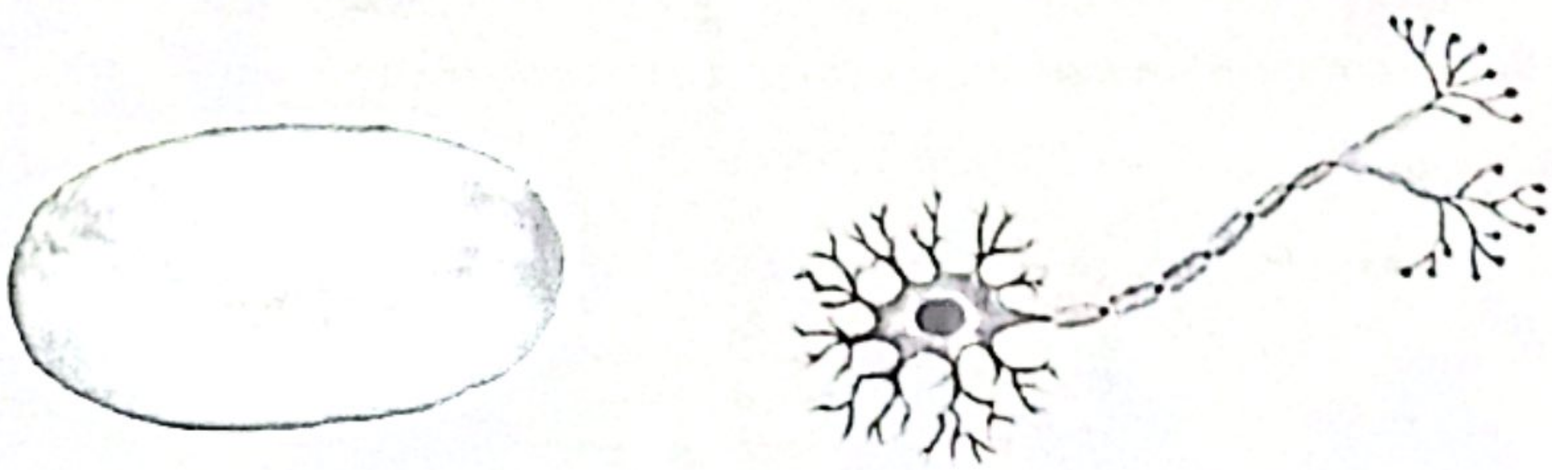
Please read and follow the instructions below:

- Please write your name, section and ID on the question paper.
- Do not turn over the paper unless instructed to do so.
- Only write in blue or black ink.
- Use of correction fluid is strictly prohibited.
- If you have any questions, raise your hand to ask the invigilator.
- Omission of working will result in loss of marks.
- Use a pencil for making straight lines.
- Answer all questions.

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1.

Fig. 6.1 shows images of cells from two different organ systems.



A B

not to scale

Fig. 6.1

(a) State the names of the cells shown in Fig. 6.1.

Plain Peer

cell A Red blood Cell (2)

cell B Cancer Cell

[2]

(b) Complete the definition of the term *tissue* by inserting the missing words.

A tissue is a group of cells with similar Structure (2) working together to perform a shared function/task.

[2]

(c) State the names of the organ systems these organs belong to.

brain Nervous System

stamen Digestive System ✗ (2)

ovary Reproduction System

[3]

[Total: 7]

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2 A student investigated the diffusion of substances through Visking tubing, an artificial membrane which has some of the properties of cell membranes.

The student made a bag of Visking tubing as shown in Fig. 4.1.

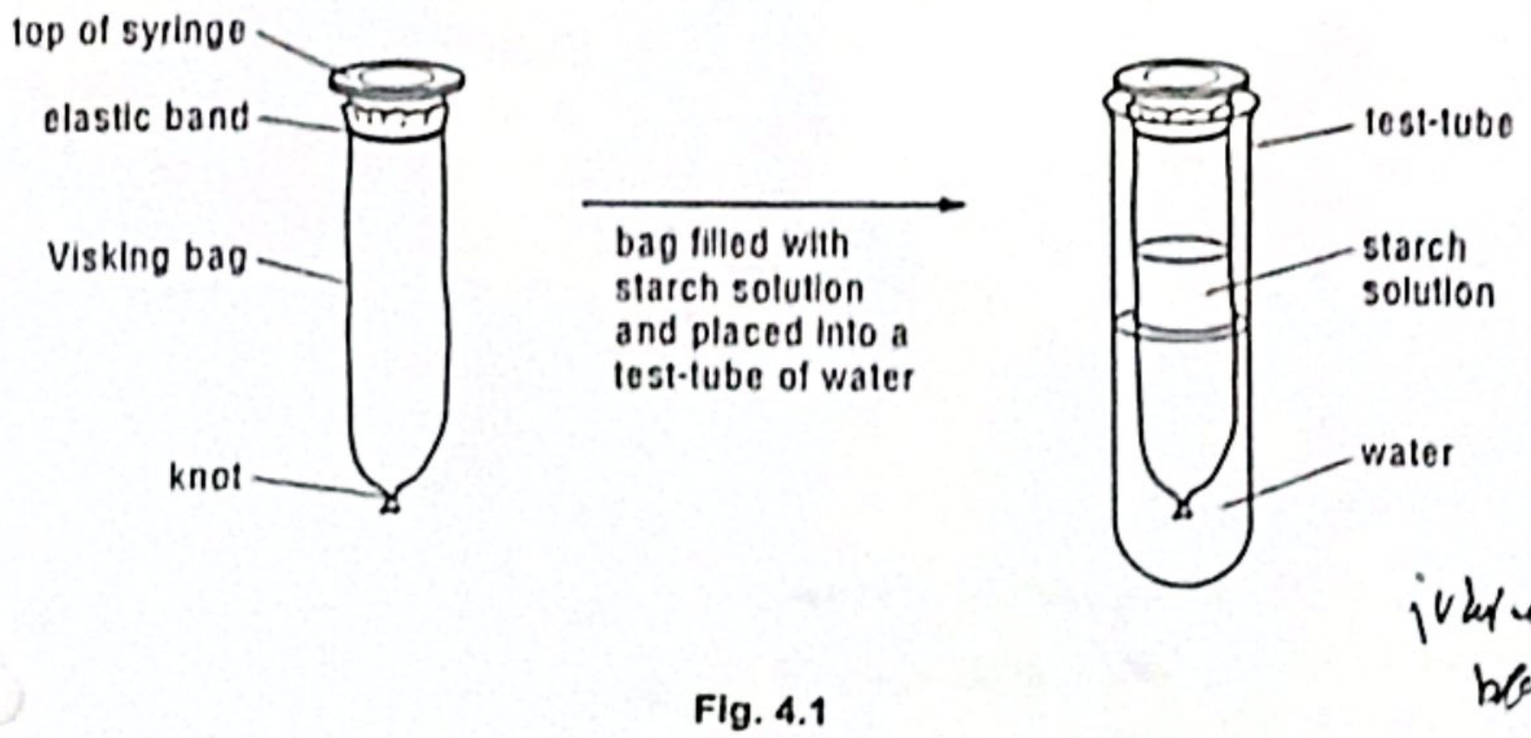


Fig. 4.1

The student added some iodine solution to the water in the test-tube.

After 30 minutes at room temperature, the contents of the Visking bag were stained blue-black, but the water outside remained a yellow colour.

(a) (i) Explain these results.

iodine molecules are small enough to pass through membrane. No starch molecules diffuse out because they are too large to pass through the membrane. Partially permeable membrane selection is to control movement of substances.

The contents inside the Visking bag were stained blue black indicating a presence of starch however outside the bag the water remained a yellow colour because osmosis happens from a region of high water potential to a region of low water potential across a partially permeable membrane so the water from outside entered into the Visking bag by osmosis but the starch solution was too large and also had less water potential so it didn't go outside in water which is why it remained yellow as didn't turn blue-black. [4]

2

DCII

DOM OSP
Water Potential

(ii) State **three** factors that influence the movement of molecules through membranes.

1 temperature

2 concentration ~~of molecul~~ gradient

3 Permeability

size, surface area

3

[3]

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3.

(a) Fig. 4.1 is a diagram of a cross-section of a root.

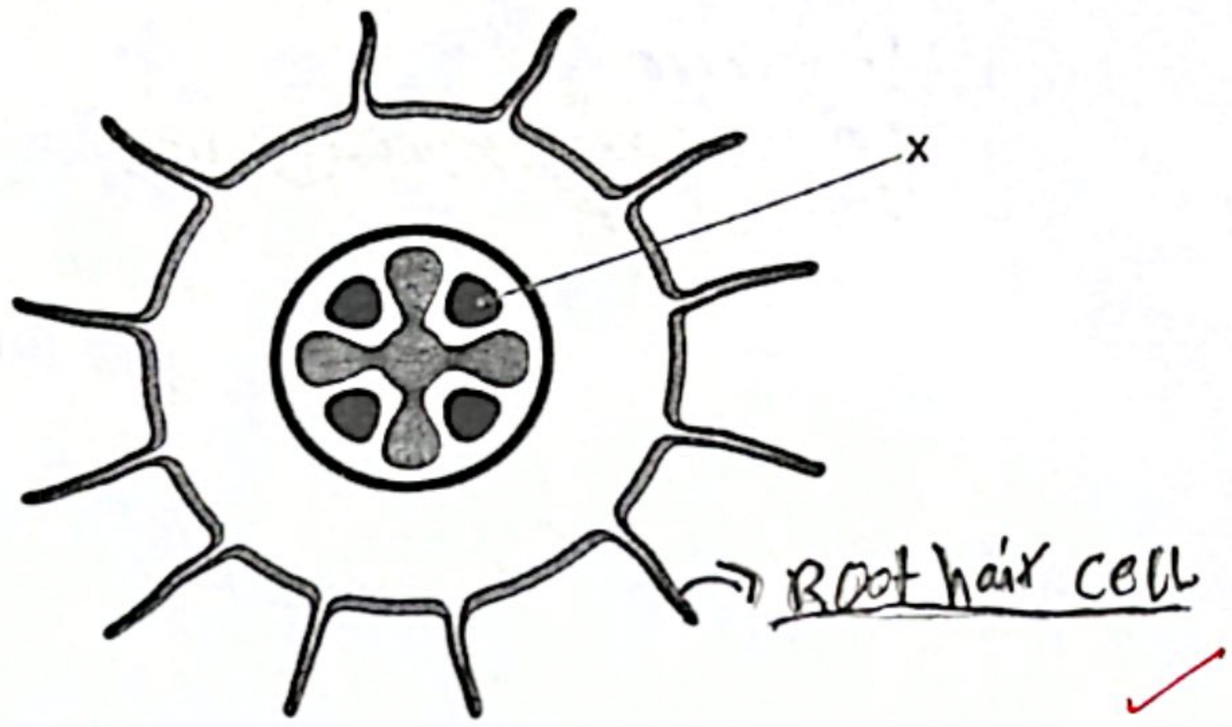


Fig. 4.1

(i) Circle two substances transported by the part labelled **X** in Fig. 4.1.

sucrose & amino acids

amino acids	cellulose	fatty acids	glucose
glycogen	starch	sucrose	

[2]

(ii) Label the part of the root in Fig. 4.1 that absorbs mineral ions from the soil with a label line and the correct name. [2]

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(b) Mineral ions are absorbed by active transport and are transported with water in the xylem.

(i) Describe what is meant by the term active transport.

It is the movement of molecules from a region of low concentration to a region of high concentration against the concentration gradient using Atp adenosine triphosphate energy from aerobic respiration and carrier proteins through a particular membrane.

3

[3]

(ii) State one function of xylem other than transport.

Stability and structural support

[1]

1

4.

(a) A balanced diet includes foods that contain calcium ions.

(i) Circle the food that has the highest calcium content.

cheese egg orange potato

[1]

(ii) Table 4.1 shows the recommended calcium intake for different age groups.

Table 4.1

age group in years	recommended calcium intake/mg per day
0–3	500
4–8	800
9–18	1300
19–50	1000
51+	1200

The list shows five statements about the data in Table 4.1.

Tick (✓) two statements that are correct descriptions of the data shown in Table 4.1.

Age group 51+ has the highest recommended daily intake of calcium.	
As age increases, the recommended daily intake of calcium decreases.	
As age increases, the recommended daily intake of calcium increases, then decreases and then increases again.	✓
The recommended daily intake of calcium doubles from ages 0–3 to ages 19–50.	✓
The recommended daily intake of calcium is higher at ages 19–50 than at ages 9–18.	

[2]

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(iii) Suggest reasons why some age groups need more calcium in their diet than others.

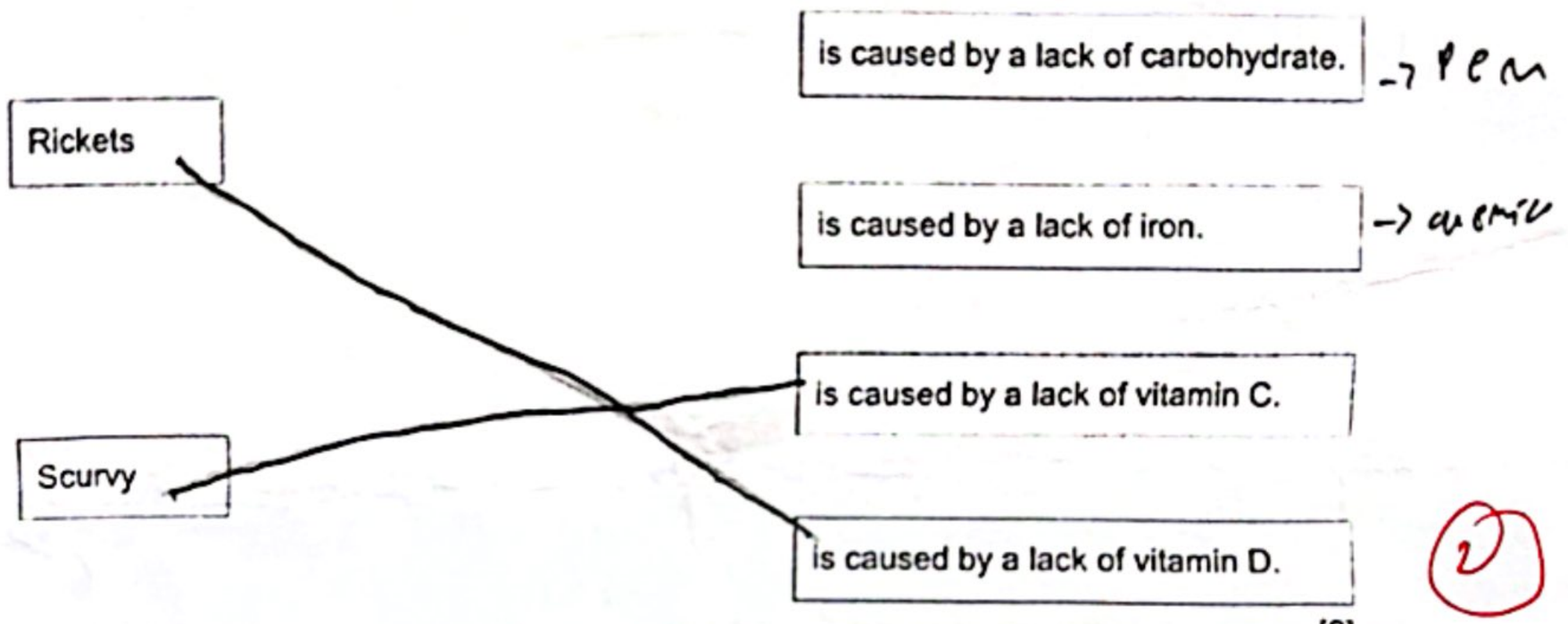
This is because they have different sizes and daily activities. Ages 0-3 need less amount because their body is small and they aren't doing alot of physical activity. 4-8 and 9-18 require more because they are in their bone growing age and ~~high amount of~~ muscle which requires alot of calcium. 19-50 require less as they have reached their adult body and stopped growing. 51+ need more because their bones are ~~keep~~ aging and weaker so they need more calcium to remain strong. [2]

(b) Some diseases are caused by an unbalanced diet.

The boxes on the left contain the names of two diseases.

The boxes on the right contain some sentence endings.

Draw **one** straight line from each box on the left to a box on the right to make **two** correct sentences.



[2]

(c) (i) State **one** dietary source of fibre.

vegetables and fruits such as cucumber. [1]

(ii) Describe the importance of fibre in the diet.

Fibre is extremely important for health of digestive system as though our body can't absorb it, it gives the intestines bulk and something to push off against which helps peristalsis and prevents constipation and colon cancer. [1]

5.

(a) Fig. 1.1 is a diagram of the digestive system.

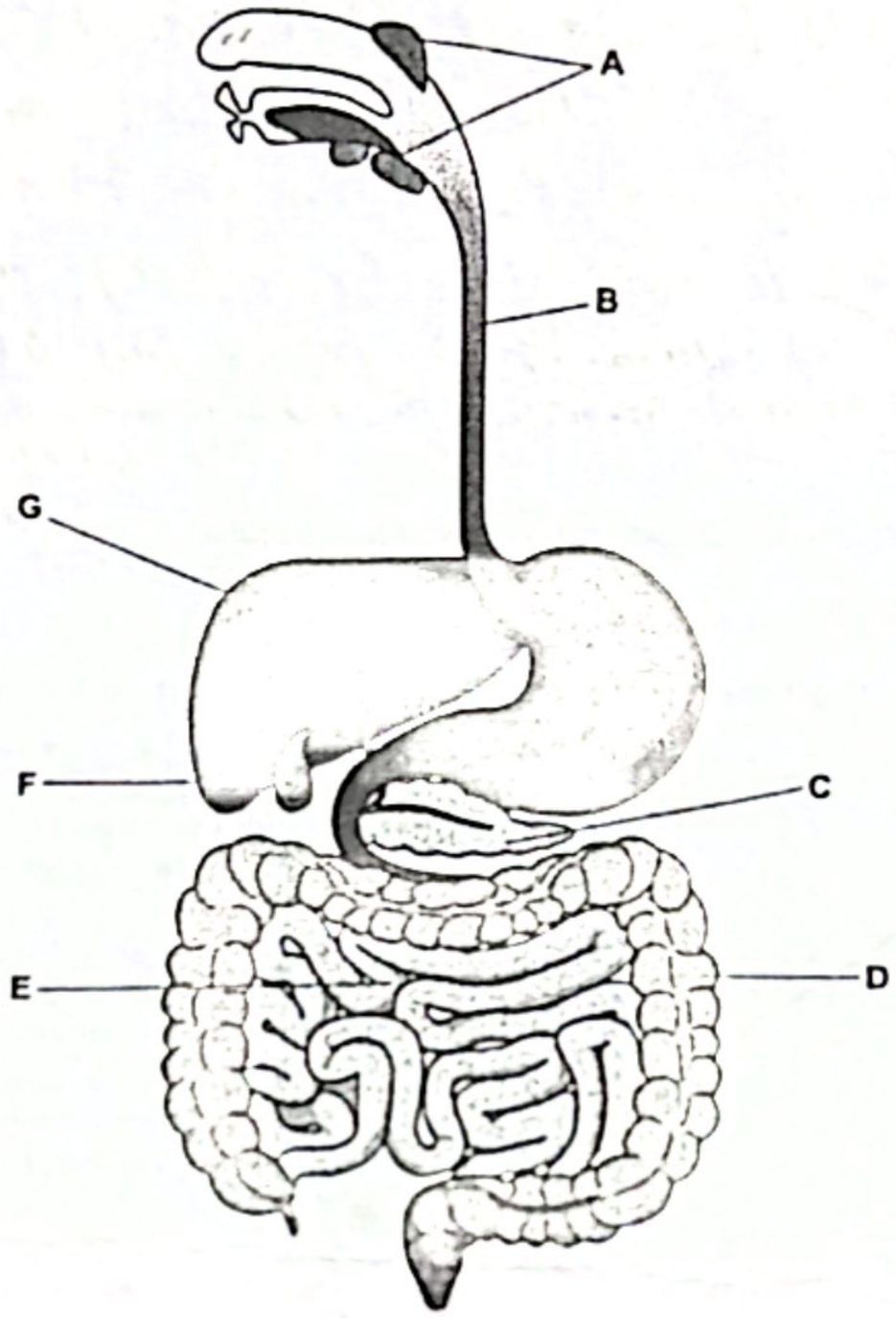


Fig. 1.1

(I) State the letter from Fig. 1.1 that represents:

where salivary amylase is produced ..A ✓..

where insulin is produced ..G ✗.. → C [Pancreas]

the liver ..G ✓..

where protease acts. ..E ✓..

3

[4]

DCCir

(II) State **one** function of the hydrochloric acid in the stomach.

~~Kill Bac~~ Ons Sachin of hydrochloric acid is to Kill harmful microorganisms and bacteria that came with the food.

1

[1]

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6.

(a) Antibodies are proteins.

State the chemical elements present in all proteins that are also found in carbohydrates and fats.

Carbon, hydrogen and oxygen. [1]

(b) Chromosomes are made of DNA.

Describe the structure of a DNA molecule.

A DNA molecule contains nucleotides made of Bases. There are 4 bases adenine, guanine, thymine and cytocin. Each base is complementary to another like adenine always pairs with thymine and cytosin always pairs with guanine. The opposite bases join together strongly. The two nucleotides wrap around eachother in a spiral ladder shape. This is known as double helix. [4]

has two strands made of Phosphate & sugar double helix structure. 4 nucleotide bases, adenine, cytocin thymin guanine